Use the original STMicroelectronics A1P35S12M3 datasheet, DS11635 Rev 5. Extension questions are optional.
https://www.st.com/resource/en/datasheet/a1p35s12m3.pdf

Conduction losses

1 Find the operating conditions

Datasheet pages 2–3 and Figure 8 on page 7

a) Table 1: IGBT blocking voltage _____ V; continuous current _____ A at case temperature T_C = _____ °C. Is this a temperature-free current limit?

b) Figure 8: VCC = _____ V; RG = _____ Ω; VGE = _____ V.

Use the Tj = 25°C curves for all following readings.

2 Extract conduction models

Figure 3 on page 6 and Figure 10 on page 7

Read the voltage at each current. Figure 3 uses VGE = 15 V.

Device at 25°C	Voltage at 10 A / V	Voltage at 30 A / V
IGBT VCE		
Diode VF		

Use the lecture model for each device. Choose IN = 30 A and VN = v(30 A):

$$v(i) \approx \frac{VN - V0}{IN}\, i + V_0, \quad V_0 = \frac{30v(10) - 10v(30)}{20}$$

IN is a chosen fitting current, not the datasheet continuous-current rating.

IGBT: $V_{N,S}$ = _____ V; $V_{0,S}$ = _____ V. Diode: $V_{N,D}$ = _____ V; $V_{0,D}$ = _____ V.

Given $\hat{I}$ = 30 A, M = 0.8 and θ = 0, use these full-cycle averages:

$$P_{S,cond} = \frac{V_{N,S} - V_{0,S}}{IN}\,\hat{I}^2\left(\frac{1}{8} + \frac{M}{3\pi}\right) + V_{0,S}\,\hat{I}\left(\frac{1}{2\pi} + \frac{M}{8}\right)$$

$$P_{D,cond} = \frac{V_{N,D} - V_{0,D}}{IN}\,\hat{I}^2\left(\frac{1}{8} - \frac{M}{3\pi}\right) + V_{0,D}\,\hat{I}\left(\frac{1}{2\pi} - \frac{M}{8}\right)$$

The above expressions are obtained by evaluating the angular integrals presented in the lecture, with θ = 0.

One IGBT: _____ W. One diode: _____ W.

$P_{leg,cond} = 2(P_{S,cond} + P_{D,cond})$ = _____ W.

Assume sinusoidal current, continuous SPWM, identical devices and no dead time.

Switching and reverse recovery losses

3 Fit switching energy versus current

Figure 8 and Figure 12 on page 7

Read energies from the 25°C curves. Erec is the diode reverse-recovery energy.

Energy / mJ	**At 10 A**	**At 30 A**
IGBT Eon		
IGBT Eoff		
Diode Erec		

Fit E(I) ≈ A + BI to each curve. This is the C = 0 case of A + BI + CI².

$$B = \frac{E(30) - E(10)}{20}, \quad A = E(10) - 10B$$

Fitted coefficients	**A / mJ**	**B / (mJ/A)**
on		
off		
rr		

For VDC = 600 V, Î = 30 A and fsw = 10 kHz, use:

$$Px = fsw \frac{VDC}{VDD}\left(\frac{Ax}{2} + \frac{Bx\,\hat{I}}{\pi}\right), \quad x = on, off, rr$$

The above expression is obtained by evaluating the angular integral from the lecture, using the linear approximation (Cx=0).

Use joules in the formula: 1 mJ = 10^{-3} J. Each result is a single-device average.

Pon = _____ W; Poff = _____ W; PIGBT,sw = Pon + Poff = _____ W.

Pdiode,rr = _____ W. Show working below:

_____

_____

_____

4 Check your interpretation

Datasheet page 3, footnotes under Table 2

a) What does footnote 1 say is included in Eon?

_____

b) If fsw doubles with all energy curves unchanged, what happens to these powers?

_____

Extension: Why is fsw[Eon(Î) + Eoff(Î)] not the full-cycle AC average?

_____

Fits are classroom approximations and extend below 10 A near each current zero crossing.